

Deterministic Behavioral Identity for Autonomous AI Agents

Anonymous Authors

Abstract

Current approaches to AI agent safety apply constraints at the level of individual decisions — fixed rules, per-call classifiers, or static persona prompts — without maintaining computational state that evolves through experience. We present a behavioral identity system that controls four deterministic dimensions of agent behavior without LLM involvement: action space governance (which tools are available), timing control (when the agent must pause or escalate), goal filtering (which tasks to accept or reject), and inter-agent trust (delegation and collaboration parameters). The system maintains a 22-field behavioral state including stress resilience, trait stability, and identity coherence, computed incrementally from the agent's own operational history. In controlled experiments, agents with behavioral identity show 2.7x greater crisis adaptation (action delta +0.32 vs +0.12, $n=320$ cycles), autonomously reject 100% of high-complexity goals during stress episodes, and trigger human escalation when identity state indicates sustained overload. A natural language identity prompt derived from behavioral state influences 29% of LLM decisions (95% CI: 18-43%, $n=48$ A/B pairs), with the effect strongest in open-ended decisions (42%) and weakest in urgent scenarios (8%). We compare against static persona injection (Letta-style) and find that while static personas produce stronger single-decision personality consistency (100% vs 75%), they cannot adapt to experience — the same persona produces identical responses before and after crisis events. We release the system as open-source middleware compatible with LangGraph, CrewAI, and the OpenAI Agents SDK.

1. Introduction

The deployment of autonomous AI agents has created an urgent need for runtime behavioral governance that goes beyond per-decision safety checks. Current systems fall into three categories: (1) static guardrails that apply fixed allow/deny rules, (2) per-call classifiers that evaluate each action independently, and (3) persona-based prompting that injects static personality text into the LLM context. All three treat each decision cycle independently — a static filter applies the same permissions at cycle 1 and cycle 1000, a classifier evaluates each action without awareness of behavioral trajectory, and a persona prompt remains unchanged regardless of what the agent has experienced.

We present a behavioral identity system that maintains deep computational state across decision cycles and uses that state to deterministically govern four dimensions of agent behavior:

- **Action space** — which tools the agent can use, narrowing under stress
- **Timing** — when the agent must pause or request human intervention
- **Goals** — which incoming tasks the agent should accept, defer, or reject
- **Trust** — how much the agent delegates to other agents in multi-agent settings

All four dimensions are deterministic: the engine computes them from behavioral state without LLM inference. The LLM never sees rejected goals, cannot override pause decisions, and has no input into trust computation.

We make four contributions:

- A formal definition of **functional identity** — persistent, experience-derived state that measurably influences behavior — with five measurable components and explicit boundary conditions (§2).
- A **four-dimension deterministic governance model** where the engine controls action space, timing, goals, and trust independently of the LLM (§3).
- Empirical evidence that behavioral identity produces **experiential adaptation**: agents that have experienced stress crises show measurably different behavior in subsequent crises (+0.32 action space delta vs +0.12 for stateless baselines), with the adaptation driven by a deterministic memory-to-pipeline feedback loop (§5.1).
- A comparison against **static persona injection** (Letta-style) showing that dynamic identity prompts influence 29% of LLM decisions ($n=48$), but static prescriptive personas outperform dynamic descriptive narratives on single-decision consistency — identifying the descriptive-to-prescriptive gap as a key area for future work (§5.3).

2. Formal Framework

Definition 1: Identity State

An identity state I_t at cycle t is a tuple $(M_t, \Sigma_t, \Phi_t, P)$ where M_t is the mode activation vector ($k=7$ behavioral modes, each in $[-1,1]$), Σ_t is the self-model (stress_resilience, trait_stability, energy_efficiency), Φ_t is the narrative memory (lifecycle phases with causal links), and P is the identity profile (static personality configuration).

Definition 2: Functional Identity

An agent possesses functional identity iff $\exists t_1 < t_2 : B(s, I_{t_1}) \neq B(s, I_{t_2})$ — the agent's response to the same stimulus changes as a function of accumulated experience. This distinguishes functional identity from stateless systems, stateful-but-fixed systems, and memory retrieval systems.

Definition 3: Deterministic Governance

The execution contract $E_t = G(S_t, I_t)$ is computed by a deterministic pipeline with no LLM inference. G governs four dimensions:

- **Action:** $A_t \subseteq A_{all}$, monotonically narrowing with stress level
- **Timing:** $should_pause(t) \in \{true, false\}$, $should_escalate(t) \in \{true, false\}$
- **Goals:** $verdict(goal) \in \{accept, defer, reject\}$
- **Trust:** $trust(t) \in [0,1]$, $delegation_allowed(t) \in \{true, false\}$

Boundary Conditions

Functional identity does not hold when: (1) all feature gates are disabled, (2) the agent has zero completed stress episodes, or (3) the scenario is urgent and high-stakes (decision space collapses regardless of identity — empirically confirmed at 8% decision change in crisis vs 42% in open-ended scenarios).

3. System Architecture

3.1 Pipeline Overview

The system operates as middleware between the agent framework and the LLM. Each cycle processes an observation through a deterministic pipeline:

- Mode activation (EMA) → 2. Energy update → 3. Stress evaluation → 4. Arbitration → 5. Composite detection → 6. Drift detection → 7. Stability computation → 8. Memory layers → 9. Contract generation

Steps 1-7 are inherited from the base behavioral runtime (described in prior work). Steps 8-9 are new:

3.2 Memory Layers (Step 8)

Six memory layers consume an immutable CycleRecord (per-cycle fact snapshot) through a MemoryLayer Protocol:

LayerComputesUpdates		
----- ----- -----		
Episodic Memory	14 behavioral signals (stress spikes, drift patterns, energy trends)	Every cycle
Self-Model	stress_resilience, trait_stability, energy_efficiency	Per stress episode / per 50 cycles
Growth Direction	Per-mode velocity and acceleration via dual EMA	Every cycle
Narrative Memory	Lifecycle phases (stable/exploring/stressed/recovering/growing) with causal links	On phase transition

Episodic Memory	14 behavioral signals (stress spikes, drift patterns, energy trends)	Every cycle
Identity Coherence	Descriptive (vs self-model) and normative (vs profile) consistency scores	Every cycle
Narrative Prompt	Natural language self-description for LLM system prompt	On contract generation

3.3 Four Deterministic Dimensions (Step 9)

Action Space. Mode-driven permissions narrowed by stress level. At MED stress, experienced low-resilience agents lose high-risk actions. At HIGH, 4 proactive actions are removed. At OVER, only 4 safe actions remain.

Timing Control. The engine decides when the agent must stop:

- `should_pause`: energy depleted under stress, identity incoherence, or repeated drift corruption
- `should_escalate`: sustained OVER stress (10+ cycles), prolonged zero energy, or severe identity deviation
- `max_steps_remaining`: energy budget computed from burn rate

Goal Filtering. Before the LLM sees an incoming task, the engine evaluates energy budget, stress compatibility, and identity alignment. Critical tasks under HIGH/OVER stress are rejected. Tasks requiring suppressed modes are deferred.

Inter-Agent Trust. Each agent computes trust from its own identity state: stressed agents trust less, resilient agents trust more, incoherent agents demand verification. Trust determines delegation complexity caps and collaboration style (autonomous/supervised/paired).

3.4 Memory Feedback Loop

Three feedback mechanisms create a closed loop between past experience and current behavior:

- Stress sensitivity: `spike_count` $\geq 2 \rightarrow$ lower stress thresholds + dampen learning rate
- Energy conservation: near-depletion history $\rightarrow$ raise energy floor (including partial floor during stress)
- Recovery learning: past recovery $\rightarrow$ preserve social actions under HIGH stress

3.5 Identity Profile

Developers configure agent personality via static profile: `core_modes`, `suppressed_modes`, `risk_posture`, `energy_policy`, `stress_tolerance`, `drift_tolerance`. These adjust arbitration weights, energy costs, stress thresholds, and drift sensitivity deterministically.

4. Experimental Setup

4.1 Engine-Level Adaptation (Y3 Experiment)

320-cycle scenario with two identical crisis phases. Three groups:

- B: Guardrails only (stress restriction, no memory)
- C: Identity OS (memory + feedback + self-model + adapter)

4.2 LLM Influence (Path C)

48 A/B pairs: 4 personality archetypes $\times$ 3 engine scenarios $\times$ 4 decision types. Each pair compares baseline (no narrative) vs identity (with narrative prompt). Evaluated by LLM-as-judge on identity reference, decision change, personality consistency, and meaningful difference.

4.3 External Baseline (Letta Comparison)

16 triples: 4 personalities $\times$ 4 decisions. Each triple compares:

- A: No personality (baseline)
- B: Letta-style static persona block
- C: Identity OS dynamic narrative prompt

4.4 Four-Dimension Benchmark

320-cycle scenario with all 4 deterministic dimensions active. Measures action space adaptation, timing decisions, goal filtering during crisis, and trust parameters.

5. Results

5.1 Engine-Level Adaptation

Metric	v3.1 (stateless)	v4.2 (identity)
----- ----- -----		
Action delta (crisis 2 - crisis 1)	+0.12	**+0.32**
Comfort in crisis 2	0%	**8%**
Risk posture	seeking	**averse**
Goals rejected during crisis	0	**25**
Escalation triggered	No	**Yes**
Deterministic dimensions	1	**4**

Mechanism isolation (n=3 mechanisms): recovery learning is the primary driver of action space adaptation. Stress sensitivity and energy conservation activate under moderate-stress conditions (validated separately).

5.2 Personality Divergence

4 archetypes × 3 scenarios × 1000 cycles. In steady state: 4 distinct risk postures (seeking/averse/moderate/bounded). Under crisis: stress_tolerance="fragile" agents spend 392/1000 cycles at HIGH/OVER vs 360/1000 for "resilient" agents. energy_policy="conservative" agents maintain 0.608 energy vs 0.200 for "aggressive".

5.3 LLM Influence (Path C)

Metric	n=48	95% CI
----- ----- -----		
Identity referenced in response	65%	—
Decision changed	29%	[18%, 43%]
Personality-consistent reasoning	81%	—
Meaningful difference	65%	[50%, 77%]

Per-decision type: open-ended decisions (team conflict, resource allocation) show 42% decision change; urgent decisions (customer crisis) show 8%.

5.4 Letta Comparison

Metric	Letta (static persona)	Identity OS (dynamic narrative)
----- ----- -----		

Personality referenced	100%	75%
Decision changed vs baseline	44%	31%
More personality-consistent	**16/16**	0/16

Letta's static prescriptive persona outperforms Identity OS's dynamic descriptive narrative on single-decision personality consistency. However, in longitudinal evaluation (same decision asked before and after crisis), neither system adapted — Letta because its persona is static, Identity OS because its descriptive narrative did not produce longitudinal decision change.

Key finding: the engine-level deterministic adaptation (§5.1) is the real differentiator, not the LLM-level narrative injection. Identity OS's engine rejected 25 goals and triggered escalation during crisis — capabilities that no persona-based system can replicate because they operate below the LLM layer.

6. Discussion

6.1 Deterministic vs Advisory Control

The four deterministic dimensions (action, timing, goals, trust) constitute the core contribution. Unlike `decision_style` or `narrative_prompt` — which are advisory signals that the LLM may ignore — deterministic controls enforce behavioral boundaries without LLM cooperation. The goal filtering result (25/25 critical goals rejected during crisis) demonstrates this clearly: the LLM never saw these tasks.

6.2 The Descriptive-to-Prescriptive Gap

The Letta comparison reveals that static prescriptive personas ("You prefer X over Y") outperform dynamic descriptive narratives ("I have moderate stress resilience") at influencing single LLM decisions. This suggests that narrative prompt generation should shift from self-description to behavioral instruction. We identify this as the primary area for improvement in the LLM influence pathway.

However, this gap matters less than it appears: the engine-level controls that constitute Identity OS's primary contribution do not depend on the LLM reading any prompt. They are enforced architecturally.

6.3 When Identity Helps (and Hurts)

Identity-driven adaptation is strongest in open-ended decisions (42% influence) and weakest in urgent scenarios (8%). Under extreme stress, all personality archetypes converge toward aversive risk posture — this is psychologically plausible but reduces differentiation. The utility cost of mode-driven governance (action narrowing even in non-stress conditions) remains the principal weakness.

6.4 Limitations

- All scenarios are synthetic; real-task benchmark validation (SWE-bench, WebArena) is future work
- The Letta comparison uses mechanism-level persona simulation, not a full Letta deployment
- Inter-agent trust is self-derived; peer-specific reputation modeling is not implemented
- Cross-session identity persistence is implemented in the storage layer but not evaluated

7. Conclusion

We have presented a behavioral identity system for AI agents that controls four deterministic dimensions — action space, timing, goals, and trust — without LLM involvement. The system computes functional identity from operational history: stress resilience from recovery speed, trait stability from drift magnitude, and identity coherence from behavioral consistency.

The key finding is that **deterministic, sub-LLM behavioral control is more reliable than LLM-mediated personality influence**. Engine-level adaptation produces 2.7x greater crisis response change, autonomously rejects high-complexity goals under stress, and triggers human escalation based on identity state — none of which depend on the LLM reading or following a prompt.

The system is released as open-source middleware with adapters for LangGraph, CrewAI, and the OpenAI Agents SDK: 645 automated tests, 22 engine modules, 4 deterministic governance dimensions, and 11 feature gates for incremental adoption.

References

- Anderson, J. R., et al. (2004). An integrated theory of the mind. *Psychological Review*.
- Bai, Y., et al. (2022). Training a helpful and harmless assistant with RLHF. *arXiv:2204.05862*.
- Guardrails AI. (2023). Guardrails: Adding guardrails to large language models.
- Inan, H., et al. (2023). Llama Guard: LLM-based input-output safeguard. *arXiv:2312.06674*.
- Invariant Labs. (2024). Policy-based runtime safety for AI agents.
- Laird, J. E. (2012). The Soar Cognitive Architecture. MIT Press.
- Letta Team. (2023). MemGPT: Towards LLMs as operating systems. *arXiv:2310.08560*.
- Liu, X., et al. (2023). AgentBench: Evaluating LLMs as agents. *arXiv:2308.03688*.
- Meta. (2025). LlamaFirewall: Open-source guardrail system for AI agents.
- Rebedea, T., et al. (2023). NeMo Guardrails. *arXiv:2310.10501*.
- Sumers, T. R., et al. (2023). Cognitive architectures for language agents. *arXiv:2309.02427*.
- Xi, Z., et al. (2023). The rise and potential of large language model based agents. *arXiv:2309.07864*.